

多元微分学测验题（二）参考答案

选择题：1. (D) 2. (C) 3. (C) 4. (B) 5. (D)

填空题：6. $2(f''_{11} - f''_{12} + f''_{22})$ 7. $\int_{-1}^0 dy \int_1^e f(x, y) dx + \int_0^1 dy \int_{e^y}^e f(x, y) dx$ 8. $\frac{\pi}{\sqrt{2}}$

9. $2\sqrt{2} - 1$ 10. $2 + \frac{\sqrt{2} + 1}{2}\pi$

解答题

$$11. I = \int_0^1 dy \int_0^{1-y} dz \int_0^{1-y-z} (1-y)e^{-(1-y-z)^2} dx = \frac{1}{4e}$$

12. 用垂直于 z 轴的平面 $z = z$ ($0 \leq z \leq t$) 截 Ω 所得截面区域为 $D_z: x^2 + y^2 \leq z$; 则

$$\iiint_{\Omega} f(z) dx dy dz = \int_0^t f(z) dz \iint_{D_z} dx dy = \pi \int_0^t z f(z) dz; \text{ 由罗比塔法则, 所求极限为 } \frac{\pi}{2} f(0).$$

13. L 的极坐标方程为 $\rho = 2 \cos \varphi$ ($-\frac{\pi}{2} \leq \varphi \leq \frac{\pi}{2}$);

$$M = \oint_L \sqrt{x^2 + y^2} ds = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} 2 \cos \varphi \cdot \sqrt{(2 \cos \varphi)^2 + (-2 \sin \varphi)^2} d\varphi = 8;$$

$$M_y = \oint_L x \sqrt{x^2 + y^2} ds = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} 4 \cos^3 \varphi \cdot \sqrt{(2 \cos \varphi)^2 + (-2 \sin \varphi)^2} d\varphi = \frac{32}{3};$$

由对称性知 $\bar{y} = 0$; $\bar{x} = \frac{M_y}{M} = \frac{4}{3}$; 曲线的质心坐标为 $(\frac{4}{3}, 0)$.

14. Γ 的参数方程为 $x = \cos \theta, y = \sin \theta, z = 1 - \cos \theta$, $\theta: 2\pi \rightarrow \pi$; 则有

$$\begin{aligned} I &= \int_{2\pi}^{\pi} [(\sin \theta + \cos \theta - 1)(-\sin \theta) + (1 - 2 \cos \theta) \cos \theta + (\cos \theta - \sin \theta) \sin \theta] d\theta \\ &= \int_{2\pi}^{\pi} (\sin \theta + \cos \theta - 2) d\theta = 2\pi + 2 \end{aligned}$$

$$15. A = \int_0^2 f(t) dt;$$

$$2I = \int_0^2 dx \int_x^2 f(x) f(y) dy + \int_0^2 dy \int_0^y f(x) f(y) dx = \int_0^2 dx \int_0^2 f(x) f(y) dy = A^2; \quad I = \frac{A^2}{2}.$$

$$16. \oint_L \frac{\partial u}{\partial n} ds = \oint_L \left[\frac{\partial u}{\partial x} \cos(\vec{n}, x) + \frac{\partial u}{\partial y} \cos(\vec{n}, y) \right] ds = \oint_L \left[\frac{\partial u}{\partial x} \cos(\vec{\tau}, y) - \frac{\partial u}{\partial y} \cos(\vec{\tau}, x) \right] ds$$

$$= \oint_L -\frac{\partial u}{\partial y} dx + \frac{\partial u}{\partial x} dy = \iint_D \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) dx dy = \iint_D \Delta u d\sigma \quad (\text{格林公式}).$$

17. 直线段 $OP: x = \xi t, y = \eta t, z = \zeta t$, $t: 0 \rightarrow 1$; 功 $W = \int_0^1 3\xi\eta\zeta t^2 dt = \xi\eta\zeta$;

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用拉格朗日乘子法求 $W = \xi \eta \zeta$ 在条件 $\frac{\xi^2}{a^2} + \frac{\eta^2}{b^2} + \frac{\zeta^2}{c^2} = 1$ 下的极值得最大值为

$$W\left(\frac{a}{\sqrt{3}}, \frac{b}{\sqrt{3}}, \frac{c}{\sqrt{3}}\right) = \frac{abc}{3\sqrt{3}}$$

$$18. \text{ 切平面方程 } \frac{xX}{2} + \frac{yY}{2} + \frac{zZ}{1} = 1; \quad \rho(x, y, z) = \frac{1}{\sqrt{\frac{x^2}{4} + \frac{y^2}{4} + z^2}};$$

$$S: z = \sqrt{1 - \frac{x^2}{2} - \frac{y^2}{2}} \text{ 在 } xoy \text{ 面的投影区域为 } D_{xy}: x^2 + y^2 \leq 2; \text{ 且}$$

$$dq = \frac{\sqrt{4-x^2-y^2}}{2\sqrt{1-\frac{x^2}{2}-\frac{y^2}{2}}} dx dy; \quad I = \frac{1}{4} \iint_{D_{xy}} (4-x^2-y^2) dx dy = \frac{1}{4} \int_0^{2\pi} d\theta \int_0^{\sqrt{2}} (4-r^2) r dr = \frac{3}{2}\pi.$$

$$19. D: x^2 + y^2 \leq h^2; V = \iint_D \left(h - \frac{x^2 + y^2}{h}\right) dx dy = \frac{\pi}{2} h^3;$$

$$S = \iint_D \sqrt{1 + \frac{4}{h^2} x^2 + \frac{4}{h^2} y^2} dx dy = \frac{\pi}{6} h^2 (5\sqrt{5} - 1); \quad \frac{dh}{dt} = -\frac{5\sqrt{5}-1}{9} \alpha;$$

$$t = \frac{9}{124\alpha} (5\sqrt{5} + 1) h_0$$

$$20. \text{ 界面由旋转抛物面 } S_1: y^2 + z^2 = 1 + x \text{ 与平面 } S_2: x = 1 \text{ 构成。} \iint_{S_2} \frac{y dx dy}{x^2 + y^2 + z^2} = 0,$$

$$\text{且由对称性知 } \iint_{S_1} \frac{y dx dy}{x^2 + y^2 + z^2} = 0; \quad S_1 \text{ 和 } S_2 \text{ 在 } yoz \text{ 面的投影域为 } D_{yz}: y^2 + z^2 \leq 2;$$

$$\iint_{S_1} \frac{dy dz}{x^2 + y^2 + z^2} = - \iint_{D_{yz}} \frac{dy dz}{(1 - y^2 - z^2)^2 + y^2 + z^2} = \frac{1}{2} \arctan 3 + \frac{\pi}{8};$$

$$\iint_{S_2} \frac{dy dz}{x^2 + y^2 + z^2} = - \iint_{D_{yz}} \frac{dy dz}{1 + y^2 + z^2} = \pi \ln 3;$$

$$\text{所以 } I = \frac{1}{2} \arctan 3 + \left(\frac{1}{8} + \ln 3\right) \pi$$

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